

# F - Avoid Division

分值: 550 分

## Problem Statement

You are given a tree with  $N$  vertices.

The vertices are numbered vertex 1, vertex 2, ..., vertex  $N$ , and the  $i$ -th edge ( $1 \leq i \leq N - 1$ ) connects vertices  $U_i$  and  $V_i$ .

给定一棵包含  $N$  个顶点的树。顶点编号为 1, 2, ...,  $N$ , 第  $i$  条边 ( $1 \leq i \leq N - 1$ ) 连接顶点  $U_i$  和  $V_i$ 。

Takahashi will color each vertex with one of colors 1, 2, ...,  $K$ .

Color  $i$  can be used to color at most  $C_i$  vertices.

高桥将为每个顶点着色, 颜色可选值为 1, 2, ...,  $K$ 。其中颜色  $i$  最多可以用于  $C_i$  个顶点。

Determine whether it is possible to color the vertices satisfying the following condition, and if so, output one valid coloring.

判断是否存在满足以下条件的着色方案, 若存在则输出任意一种合法方案。

- For every edge, there exists some  $i$  ( $1 \leq i \leq K$ ) such that both of the two subtrees obtained by cutting the tree at that edge contain at least one vertex colored with color  $i$ .
- 对于每条边, 存在某个  $i$  ( $1 \leq i \leq K$ ), 使得将该边删去后得到的两个子树中, 都至少有一个顶点被染为颜色  $i$ 。

You are given  $T$  test cases; solve each of them.

共有  $T$  个测试用例, 你需要分别处理每个测试用例。

## Constraints

- $1 \leq T \leq 10^5$
- $2 \leq N \leq 3 \times 10^5$
- $1 \leq K \leq N$

- $1 \leq U_i, V_i \leq N$
- The given graph is a tree.
- $1 \leq C_i \leq N$
- $C_1 + C_2 + \cdots + C_K \geq N$
- All input values are integers.
- The sum of  $N$  over all test cases does not exceed  $3 \times 10^5$ .
- $1 \leq T \leq 10^5$
- $2 \leq N \leq 3 \times 10^5$
- $1 \leq K \leq N$
- $1 \leq U_i, V_i \leq N$
- 给定的图是一棵树。
- $1 \leq C_i \leq N$
- $C_1 + C_2 + \cdots + C_K \geq N$
- 所有输入值均为整数。
- 所有测试用例的  $N$  之和不超过  $3 \times 10^5$ 。

## Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
T
case1
case2
⋮
caseT
```

case<sub>*i*</sub> represents the *i*-th test case. Each test case is given in the following format:

case<sub>*i*</sub> 表示第 *i* 个测试用例。每个测试用例按照以下格式给出：

$N$   $K$   
 $U_1$   $V_1$   
 $U_2$   $V_2$   
 $\vdots$   
 $U_{N-1}$   $V_{N-1}$   
 $C_1$   $C_2 \dots C_K$

---

## Output

Output  $T$  lines.

On the  $i$ -th line ( $1 \leq i \leq T$ ), output the answer for the  $i$ -th test case as follows.

输出  $T$  行。在第  $i$  行 ( $1 \leq i \leq T$ ) 中, 按如下格式输出第  $i$  个测试用例的答案。

If it is impossible to color the vertices satisfying the condition, output  $-1$  alone.

Otherwise, output  $N$  integers  $X_1, X_2, \dots, X_N$  ( $1 \leq X_i \leq K$ ) separated by spaces, such that coloring vertex  $i$  with color  $X_i$  for  $i = 1, 2, \dots, N$  satisfies the condition.

如果不存在满足条件的着色方案, 单独输出  $-1$ 。否则, 输出  $N$  个整数  $X_1, X_2, \dots, X_N$  ( $1 \leq X_i \leq K$ ), 用空格分隔, 其中对于每个  $i = 1, 2, \dots, N$ , 顶点  $i$  被染为颜色  $X_i$ , 这样的着色方案满足条件。

## Sample Input 1

```
2
5 3
1 2
1 3
2 4
2 5
2 2 2
3 3
1 2
2 3
1 1 1
```

## Sample Output 1

```
3 2 2 1 1
-1
```

For the first test case, suppose the vertices are colored as in the sample output. That is, vertex 1 is colored with color 3, vertex 2 with color 2, vertex 3 with color 2, vertex 4 with color 1, and vertex 5 with color 1.

对于第一个测试用例，假设顶点按样例输出的方式着色。即顶点 1 染为颜色 3，顶点 2 染为颜色 2，顶点 3 染为颜色 2，顶点 4 染为颜色 1，顶点 5 染为颜色 1。

If we cut the edge connecting vertices 1 and 2, the tree splits into the subtree consisting of vertices 1, 3 and the subtree consisting of vertices 2, 4, 5; both subtrees contain a vertex colored with color 2 (vertex 3 or vertex 2, respectively).

Such a color exists no matter which edge of the given tree is cut, so the coloring in the sample output satisfies the condition.

如果我们删去连接顶点 1 和 2 的边，树会被拆分为由顶点 1, 3 组成的子树和由顶点 2, 4, 5 组成的子树；两个子树中都包含颜色为 2 的顶点（分别是顶点 3 和顶点 2）。无论删去给定树的哪一条边，都存在这样的颜色，因此样例输出的着色方案满足条件。

For the second test case, there is no valid coloring satisfying the condition.

对于第二个测试用例，不存在满足条件的合法着色方案。